

Applied Math

Complete Lecture Notes

PAGE 1

0. Statistics; Numerical Characteristics; Parameter Estimation, Hypothesis Testing

1. - Mean = $\text{mean}(x)$ - Median = $\text{median}(X)$ - Range = $\text{range}(X)$ - Correlation Coefficient = $\text{corrcoef}(x, y)$; Covariance Matrix = $\text{cov}(x, y)$ - Variance = $\text{var}(X)$, Standard Deviation = $\text{std}(X)$ - Sample Standard Deviation = $\text{stdev}(x, 1)$
 1. - $[\mu_0, \sigma^2] = \dots \text{data}(\text{param})$; Random Number = $\dots \text{rand}(\text{param}, \text{count})$ - $f(x_0) = \dots \text{pdf}(X, \sigma^2)$; $F(x_0) = \dots \text{cdf}(X_a, \text{param})$ - $X_0 = \dots \text{inv}(P^2(0), \text{param})$
 1. Normal Distribution Parameter Estimation - $[\mu_0, \sigma^2] = \text{normfit}(X)$
 1. Normal Distribution Hypothesis Testing ($h=0$: Accept; $h \neq 1$: Reject) - Single μ_0 * $z = (\bar{x} - \mu_0) / \sigma / \sqrt{n} \sim N(0, 1)$ - Two-tailed: $|z| \leq U_{1-\alpha/2}$ * Single-sided right tail test: $-z \geq u_\alpha / U_{1-\alpha}$ * Single μ_0 (σ^2 unknown) - $t = \bar{x} / s / \sqrt{n} \sim t(n-1)$ * Two-tailed: $|t| \leq t_{1-\alpha/2}$ - Single-sided right tail test: $* t \geq t_\alpha / t_{1-\alpha}$ **Single σ^2 ** - $\chi^2 = (n-1)S^2 / \bar{S}^2 \sim \chi^2(n)$ - Two-tailed: $\chi^2 \leq x_{1-\alpha/2}$ * Single-sided right tail test: $- \chi^2 \geq x_{1-\alpha}$ **Two μ_0 (σ^1, σ^2 known) ** - $z = \bar{x} - y / \sqrt{(\sigma_1^2/n + \sigma_2^2/n)} \sim N(0, 1)$ - Two-tailed: $|z| \leq U_{1-\alpha/2}$ * Single-sided right tail test: $-z \geq u_\alpha / U_{1-\alpha}$ **Two μ_0 (σ^1, σ^2 unknown) ** - $t = \bar{x} - y / \sqrt{(S_1^2/n + S_2^2/n)} \sim t(n-1, n)$ - Two-tailed: $|t| \leq t_{1-\alpha/2}$ * Single-sided right tail test: $-t \geq t_\alpha / t_{1-\alpha}$ **Two σ^2 ** * $F = S_1^2 / S_2^2 \sim F(n-1, n)$ - Two-tailed: $s_1 \leq S_3^2$ * Single-sided right tail test: $-F \geq f_{1-\alpha/2}$ **Z-test** * $z = (\bar{x} - p_0) / \sqrt{(p_0(1-p_0)/n)}$ * Two-tailed: $|z| \leq U_{1-\alpha/2}$ - Single-sided right tail test: $* z \geq u_\alpha / U_{1-\alpha}$ ** χ^2 -test** - $\chi^2 = (n-p) / p(1-p)$ * Two-tailed: $|z| \leq U_{1-\alpha/2}$ - Single-sided right tail test: $* z \geq u_\alpha / U_{1-\alpha}$
-

Mathematical Interpolation and Numerical Integration

1. Lagrange Polynomial Interpolation:

$$y = \text{lagr}(x_0, x)$$

2. Piecewise Linear Interpolation:

$$y = \text{interp1}(x_0, x)$$

3. Cubic Spline Interpolation:

$$y = \text{spline}(x_0, x)$$

4. Trapezoidal Integration:

$$S = \text{trapz}(x, y)$$

5. Quadrature Integration:

$$S = \text{quad}(f, a, b)$$

6. Monte Carlo Integration:

$$S = \frac{m}{n} A,$$

7. Mean Value Integration:

$$S = \frac{1}{n} A,$$

8. Numerical Differentiation and Ordinary Differential Equations

****1st Order Forward Difference:****

$$y' = \text{diff}(y)$$

****2nd Order Forward Difference:****

$$y' = \text{diff}(y)$$

1. Euler's Method: - Forward: Local error $O(h^2)$ 2. Backward Euler's Method 3. Central Difference Formula: - Local error $O(h^3)$ 4. Improved Euler's Method 5. 3rd Order Runge-Kutta: 6. 4th Order Runge-Kutta:

Linear Systems of Equations and Linear Differential Equations

1. Direct Method (Small-Scale):

$$x = A \backslash b,$$

2. Cholesky Decomposition:

3. Symmetric Positive Definite Matrix:

****Iterative Methods (Large-Scale, Ill-Conditioned)****

$$x^{(k+1)} = Bx^k + f,$$

$$\|B\| < 1 \Leftrightarrow \text{convergence}.$$

$$[L, U] = \text{lu}(A),$$

****Matrix Decomposition:**** - Cholesky:

$$L^T = \text{chol}(A).$$

****Eigenvalue Decomposition:****

$$\rho(B) = \max(\text{abs}(\text{eig}(B))).$$

PAGE 3

Mathematical Methods in Numerical Analysis

1) Direct Iteration: $B = D^{-1}(L + U)$, $f = D^{-1}b$ $[x, \text{time}, B] = J(A, b, x_0, \text{tol})$ Abs: $|a_{ij}| > \sum_{i \neq j} a_{ij}$

2) Gauss-Seidel Iteration: $B = (D - L)^{-1}U$, $f = (D - L)^{-1}b$ $[x, \text{time}, B] = G(A, b, x_0, \text{tol})$ Abs:

$|a_{ii}| > \sum_{i \neq j} a_{ij}$ 3) SOR Iteration: $B = B(w)$, $f = f(w)$ Abs: $0 < w \leq 1$, stable

Note: $A = D - L - U$ $D = \text{diag}(\text{diag}(A))$, $U = -\text{triu}(A, 1)$, $L = -\text{tril}(A, -1)$

Norm: norm(x,p) Condition Number: cond(x) ≥ 0 Cond > 1 then ill-conditioned

Sparse Matrix: S = sparse(i,j,val,m,n); SS=full(S)

Nonlinear Equations (System): $x^{(k+1)} = \phi(x^{(k)})$, until $|\phi^{(p)}(x^*)| < \text{tol}$ [x, time] =

newton('f','df',x0,n,tol) Single Root: [x, f] = fsolve(@f,x0,opt,") Multiple Roots:

root=roots(coefficients)

1. Newton (Iteration) Method: $\phi(x_k) = x_{k+1} - f(x_k)/f'(x_k)$

2. Newton (Quotient) Method: $\phi(x_k) = x_{k+1} - f(x_k)/f'(x_{k-1})$

3. Quasi Newton (Iteration) Method:

4. Quasi Newton (Quotient) Method:

1D Root Finding: 2. Unconstrained Optimization: 3. Constrained Optimization (using gradient)

[x, f] = fminbnd(@f,v0,v1,opt,") [x,fv]=fminunc(@f,x0, opt,") [x fv] = fminsearch(@f, x0,opt,")

Roots: Find the roots of the derivative equation.

PAGE 4

Mathematics Assignment Paper

1. Multivariate regression; polynomial linear regression, multivariable quadratic regression and non-linear regression.

1. Residuals: $\hat{e}_i = e_i = y_i - \hat{y}_i$

1. Residual sum of squares: $Q = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (e_i^2)$

1. Regression sum of squares: $u = \sum_{i=1}^n (y_i - \hat{y})^2$

1. Total sum of squares: $S = \sum_{i=1}^n (y_i - \hat{y})^2 = u + Q$

1. Residual sum of squares: $\hat{\sigma}^2 = S^2 = \frac{Q}{n-m-1}$, m is the number of parameters; $n - m - 1$: degrees of freedom

1. Determination coefficient $R^2 = \frac{u}{S} \in (0, 1)$

Linear Regression

$$\beta = (\Phi^T(x) \cdot \phi(x)) / (\Phi^T(x) \cdot y), \text{ prediction interval } \approx [y_0 - u_{1-\alpha/2} s, y_0 + u_{1-\frac{\alpha}{2}} s]$$

Multivariate Linear Regression

$$[B, pci, e, eci, [R^2, F, pF(1, n-m-1), Fval], s^{\{2\}}] = \text{regress}(y, x, \alpha)$$

Polynomial Linear Regression:

1. Polynomial regression

$$p = \text{polyfit}(x, y, m) \quad B, pCI, e \leftarrow \text{polytool}(x, y, m, a)$$

1. Multivariable Quadratic Regression: Ascending

$$B, s, e \leftarrow \text{rstd}(x, y, \text{'model'}, a)$$

Non-Linear Regression

1. $[B, \text{residual sum of squares}, \text{residuals}] = \text{lsqnonlin}(@F, B0, v1, v2, \text{opt}, \text{param})$ Residuals = $F(B, \text{param})$
1. $[B, \text{residual sum of squares}, \text{residuals}] = \text{lsqcurvefit}(@F, B0, x, y, v1, v2, \text{opt}, \text{param})$
 $y = F(B, x)$

All of the above are based on least squares principle and thus belong to unconstrained optimization.

PAGE 5

Linear Programming; Nonlinear Programming

1. Linear programming:

$$\min z = c^T x$$

2. Quadratic Programming:

$$\min z = 0.5x^T Hx + c^T x$$

3. Nonlinear Programming:

$$\min z = f(x)$$
